

To: Technical Manager
From: Josh Tsutaoka
Re: Fake News Detection Model
Date: March 2026

Summary

I built a binary text classifier to identify fake news articles using the Kaggle Fake and Real News dataset containing around 45,000 articles. After testing three models; Logistic Regression, Linear SVM, and Naive Bayes all with TF-IDF and Counter Vectorization; I would recommend Logistic Regression with TF-IDF for deployment

Why This Model?

Weight F1 was the priority metric for fake news detection because errors in both directions carry consequences. Missed detections allow misinformation to reach readers, while false alarms mean legitimate journalism gets suppressed. Fake news detection requires balance across both error types. Logistic Regression with TF-IDF achieved a 0.9976 F1 on the Fake class and 0.9973 on the Real class after tuning, outperforming Naive Bayes which came in at 0.9650 and 0.9626. It also provides direct visibility into which terms are driving predictions, making it easier to audit and explain decisions to stakeholders.

Key Limitation

The model likely learned source writing style rather than genuine misinformation signals. Real articles in the training data are almost entirely formal wire copy while fake articles come from blogs, meaning that the model may be detecting formatting patterns rather than factual accuracy. This showed up when testing on fresh examples outside of the training data and testing data. The accuracy dropped to 65%, I recommend retraining on more stylistically diverse data and monitoring out-of-distribution articles before any production deployment.